CS383 · SOFTWARE ENGINEERING

Midterm Model 1 Practice

20 marks

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01 Questions

Q1 MCQ · 2 PTS

Multiple choice questions

1. "Software as a Service (SaaS)" slides list one benefit as pay-as-you-go and accessibility, and one drawback as:
   - ☐ Guaranteed performance
   - ☐ Full developer control
   - ☐ Security/trust vulnerabilities and internet dependence
   - ☐ Free on-prem hardware
2. According to "Application types," what determines which engineering methods/tools to use?
   - ☐ Project manager's personal preference
   - ☐ Application type, customer requirements, team background, and tools
   - ☐ Software cost alone
   - ☐ All of the above
3. A key weakness of the Waterfall model in the slides is:
   - ☐ Low documentation
   - ☐ Iterative
   - ☐ Lack of any structure
   - ☐ Inflexible to changing requirements and late problem discovery
4. To cope with change, there are two approaches
   - ☐ Code-and-fix & Big-bang integration
   - ☐ Change prevention & big-bang integration
   - ☐ Prototyping & incremental delivery
   - ☐ Waterfall with extended design phase & late verification

Q2 USER / SYSTEM · FR / NFR · 4 PTS

For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR).

Requirement	User vs. System	FR vs. NFR
"The system should be easy to use for first-year students."		
"The application shall allow password reset using a verified email."		
"The report generation shall be fast."		
"The interface will display student GPA to two decimal places."		

Q3 SPRINT CALCULATIONS · 4 PTS

Team of 6 people; workday = 8h. Last 4 sprints completed story points: 22, 28, 26, 24.

1. Compute the sprint velocity (use the average sprint story points). (1pts)
2. Compute focus factor (SprintVelocity ÷ (people × hours per day)). (1pts)
3. For a 3-week sprint (Mon–Fri, weekends off), compute effective team capacity (hours) using the focus factor. Show formulas and round sensibly. (1pts)
4. If the next sprint backlog is 30 story points, state whether the plan is feasible and why. (1pts)

   Feasibility: ☐ Feasible or ☐ Prudent Justification:

Q4.1 PROCESS MODEL · 4 PTS

For each project, choose one development process model: Agile, Plan-Driven, or Component-Based.

Project 1 — Smart-City Sensor Hub

• Stakeholder Presence: Moderate; city IT holds bi-weekly reviews; end-users (operators) rarely available.

• Requirement Volatility: Medium; new sensor types and vendors appear mid-project, but core data collection goals are stable.

• Project Size & Team: 20 engineers across 3 vendors; distributed.

• Architecture Constraints: Must integrate **COTS gateways** and vendor SDKs; strict uptime (99.9%).

• Compliance: Must pass basic cybersecurity hardening but no heavy certification.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q4.2 PROJECT 2

Project 2 — Banking Regulatory Reporting

• Stakeholder Presence: Limited; auditors and compliance office set requirements up front and review only at stage gates.

• Requirement Volatility: Low; templates and KPIs fixed by regulation for the next 3 years.

• Project Size & Team: 30 people, multi-department; formal PMO.

• Documentation & Traceability: Detailed SRS and SDD, audit trails, sign-offs required.

• Risk Profile: High penalty for defects; schedule predictable.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q4.3 PROJECT 3

Project 3 — Campus Social-Learning MVP

• Stakeholder Presence: High; product owner and pilot students give **daily** feedback.

• Requirement Volatility: High; features frequently reprioritized to find product-market fit.

• Project Size & Team: 6 developers, co-located; CI and CD in place.

• Time to Market: Need public beta in **4–6 weeks**; success measured by adoption and engagement.

• Architecture: Greenfield; minimal legacy constraints.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q4.4 PROJECT 4

Project 4 — Medical Device Companion App

• Stakeholder Presence: Moderate; clinicians available for periodic usability reviews; regulatory team sets constraints.

• Requirement Volatility: Medium; UI iterations expected, but clinical workflows and data formats are fixed by standards.

• Project Size & Team: 15 people; UX + mobile + back-end.

• Compliance: Must align with **IEC 62304** software life-cycle documentation and **ISO 14971** risk management; app is Class II accessory.

• Integration: Uses a certified SDK from the device vendor for data acquisition.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q5 PRODUCT / ORGANIZATION / EXTERNAL · 6 PTS

For each statement, indicate the requirement type: Product, Organization or External

1. The API shall respond in ≤ 300 ms (p95) for read requests under 100 concurrent users.

   Choose: ☐ Product ☐ Organization ☐ External

2. All new services must be built in Java 21 and deployed on the organization's Kubernetes platform.

   Choose: ☐ Product ☐ Organization ☐ External

3. Patient data must comply with HIPAA and local privacy law X.

   Choose: ☐ Product ☐ Organization ☐ External

4. The mobile app shall support offline read-only access for 24 hours.

   Choose: ☐ Product ☐ Organization ☐ External

5. The system shall export reports in CSV and PDF.

   Choose: ☐ Product ☐ Organization ☐ External

6. Device firmware must interoperate with ISO/IEC 14443 Type A NFC readers.

   Choose: ☐ Product ☐ Organization ☐ External